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Enhancing Consent Policy Privacy in Blockchain-Based Healthcare Data Sharing

Abstract

Blockchain-based consent management makes healthcare data sharing auditable, but public-chain enforcement can reveal the consent policies. Their conditions, such as disease-specific research restrictions, can disclose sensitive information about the data subject. To address this, we propose the Trusted Proof Service (TPS)-assisted Zero-Knowledge Policy Interpreter (TPS-ZKPI), in which the TPS generates zero-knowledge proofs of access decisions that a smart contract verifies against committed policies. We compile each policy into a fixed-length program for a reusable zero-knowledge policy interpreter, so all policies share one verification interface, and mask residual gas differences with randomized gas padding. We evaluate TPS-ZKPI on Polygon Amoy with supervised and unsupervised attacks by a passive observer inferring hidden policy properties from public-chain metadata. The results show no leakage in 43 of 45 supervised tasks under a criterion requiring mean AUC above 0.55 and significance in at least half of 30 splits, and unsupervised clustering finds no policy structure; the few exceptions trace to chance correlations with the padding. Furthermore, an ablation attributes the protection to randomized gas padding, without which the same attacks often recover policy content and structure. Overall, our empirical results show that TPS-ZKPI resists the modeled metadata attacks and provides auditable consent enforcement on a public blockchain.

Keywords: blockchain; consent policy privacy; zero-knowledge proofs; Data Use Ontology (DUO); healthcare data sharing; metadata leakage

1. Introduction

Blockchain technology has emerged as a promising foundation for decentralized and auditable healthcare data sharing [1–3]. Its transparency and immutability allow consent enforcement and authorization decisions to be verified and traced across institutions. However, the same transparency means that consent enforcement on a public blockchain can expose information about the governing consent policies, even when the health data themselves remain encrypted [4,5].

This exposure matters because consent policies in healthcare are often privacy-sensitive [6]. A *consent policy* encodes the conditions under which a data subject authorizes others to access or use their health data, such as who may access the data, for what purpose, and under what constraints. For example, a policy based on biomedical data-use terms may reveal whether a dataset is restricted to disease-specific research, whether ethics approval is required, or whether access is limited to particular institutions or projects. If such a policy is written into a smart contract, embedded in verification logic, or reflected in public execution traces, unauthorized observers may infer information about the protected resource, the data subject's constraints, or the requester's authorization context.

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Existing privacy-preserving blockchain solutions do not address this problem, because they mainly protect data contents, user identities, or transaction details [7–9]. Encrypted data-sharing mechanisms such as attribute-based encryption (ABE) and proxy re-encryption (PRE) control access to data [10,11], and zero-knowledge transaction systems hide asset ownership, values, or participant relationships [12]; none of them protects the confidentiality of the policies that govern access decisions. Hidden-policy ABE [13,14] does hide policy content, but it does not control what a public chain exposes while it enforces the policy. On a public chain, a policy can leak not only through explicit disclosure but also through metadata: the size of the calldata, the layout of the public inputs, the emitted events, the verifier interface, and the gas consumed. Policy-private consent enforcement therefore requires more than data confidentiality and zero-knowledge proofs of correct enforcement. It also requires limiting what the public-chain transcript reveals about the hidden policy.

This paper studies policy-private consent enforcement on public blockchains for healthcare data sharing. The challenge is to let an on-chain verifier check that a requester satisfies a provider-defined consent policy, while ensuring that observers who see only public-chain metadata learn minimal information about the governing policy and the requester's authorization evidence. We focus on Boolean consent policies, including policies derived from biomedical data-use conditions such as terms from the Data Use Ontology (DUO), the Global Alliance for Genomics and Health (GA4GH) standard vocabulary of consent-based data-use conditions [6], and disease-scope constraints [15].

To verify policy satisfaction on-chain without exposing the policy in the public-chain transcript, we propose the Trusted Proof Service-assisted Zero-Knowledge Policy Interpreter (TPS-ZKPI). In TPS-ZKPI, the data provider commits to a hidden consent policy on-chain and provisions the corresponding policy-bound proving material to a Trusted Proof Service (TPS). A requester submits its policy-relevant authorization evidence, such as attributes, credentials, and access context, to the TPS, which constructs the private witness and generates a zero-knowledge proof that the requester satisfies the committed policy. A smart contract then verifies this proof against the provider's policy commitment. As a result, every authorization decision remains publicly auditable, while the policy logic never appears in the public-chain transcript.

To prevent the transcript itself from revealing the policy, TPS-ZKPI combines four design elements. First, proof generation at the TPS keeps confidentiality-sensitive policy material and requester evidence off the chain. Second, heterogeneous Boolean consent policies are compiled into fixed-length Reverse Polish Notation (RPN) programs, so that every policy has the same length. Third, building on the standard commit-and-prove paradigm, a reusable zero-knowledge policy interpreter evaluates these programs; this avoids policy-specific verifier contracts and allows policies to evolve without changing the public verification interface. Fourth, transcript normalization, including randomized gas padding, reduces the remaining policy-dependent variation in calldata, event logs, public parameters, and gas consumption.

We evaluate TPS-ZKPI against a passive public-chain observer who sees only the public-chain transcript. The evaluation expresses consent policy privacy as three objectives: Policy Content Hiding (PCH), Policy Indistinguishability (PI), and Structure and Parameter Privacy (SPP). For these objectives, supervised and unsupervised attacks test whether the public-chain metadata allow the observer to recover hidden policy content, to distinguish otherwise-equivalent policies, or to infer structural and parameter properties of a policy. The results show no leakage in 43 of the 45 supervised tasks under our leakage criterion, which flags a task only when its mean area under the receiver operating characteristic curve (AUC) exceeds 0.55 and the attack is statistically significant in at least half

of 30 policy-disjoint splits; the other two trace to chance correlations with the randomly drawn padding targets. Unsupervised clustering likewise finds no policy structure beyond the same padding effect. Furthermore, a gas-padding ablation shows that, without padding, the same attacks recover a policy-correlated gas signal, which randomized gas padding suppresses.

This paper makes the following contributions:

- We propose TPS-ZKPI, a TPS-assisted zero-knowledge approach to publicly verifiable and auditable consent enforcement. The soundness of on-chain verification and the auditability of submitted decisions hold without trusting the TPS, whereas policy confidentiality, faithful witness construction, and liveness rely on trusting the TPS.
- We encode each Boolean consent policy as a fixed-length RPN program over a bounded token alphabet and evaluate it in a single zero-knowledge circuit. Combined with policy commitments, this approach provides a policy-independent on-chain verification interface, allowing policies to evolve without changing the on-chain verifier.
- We empirically evaluate how much policy information remains in the public-chain metadata on the Polygon Amoy testnet[1] using supervised and unsupervised attacks, and quantify the gas overhead of the randomized gas padding that suppresses the remaining gas channel.

The remainder of this paper is organized as follows. Section 2 reviews the technical foundations: cryptographic access control in blockchain healthcare, zero-knowledge proofs, and the Data Use Ontology. Section 3 discusses related work and positions TPS-ZKPI against existing blockchain consent-management, privacy-preserving blockchain, hidden-policy, private policy-evaluation, reusable proving, and metadata-leakage approaches. Section 4 defines the system and trust model, the public-chain adversary, and the privacy objectives. Section 5 presents the TPS-ZKPI design. Section 6 reports the empirical evaluation. Section 7 discusses limitations and future work, and Section 8 concludes the paper.

2. Background

This section reviews the technical foundations relevant to policy-private consent enforcement on blockchains. We first introduce the cryptographic access-control mechanisms commonly used in blockchain-based healthcare data sharing. We then review zero-knowledge proofs and DUO [16], which together provide the basis for the policy representation and verification mechanisms used in our approach.

2.1. Cryptographic Access Control in Blockchain Healthcare

Fine-grained consent management in blockchain-based healthcare data sharing is commonly built on cryptographic access-control primitives, chiefly ciphertext-policy ABE (CP-ABE) and PRE [11,17,18]. In CP-ABE, the encrypting party embeds an attribute-based access policy into the ciphertext, so that only requesters whose credentials satisfy the policy can decrypt the protected data [10]. PRE lets a semi-trusted intermediary transform a ciphertext encrypted for one party into one decryptable by a designated requester, delegating access without exposing the plaintext [18]. Both primitives remain the building blocks of current blockchain healthcare designs; for example, recent schemes combine PRE with on-chain records to delegate access to medical records and Internet of Medical Things (IoMT) data [19,20]. These primitives determine who may obtain the data. However, stan-

[1] https://amoy.polygonscan.com/txs?a=0x461853d10fB9A286a75dE05c9B166e77645160D7

dard CP-ABE does not keep the access policy confidential because the policy is attached to the ciphertext in plaintext.

2.2. Zero-Knowledge Proofs

Zero-knowledge proofs are cryptographic protocols that allow a prover to convince a verifier that a statement is true without revealing any information beyond the validity of the statement [21]. Zero-knowledge proof systems support verifiable computation over private inputs, so that complex predicates can be proven without disclosing the underlying data. In blockchain settings, zero-knowledge proofs are widely used to combine public verifiability with privacy. For example, Zerocash and Tornado Cash use these proofs to conceal transaction details or provide transaction anonymity [9,22]. These properties make zero-knowledge proofs a natural building block for proving that a requester satisfies a hidden consent policy, which is the direction our approach develops.

2.3. The Data Use Ontology

DUO provides a standardized vocabulary for expressing biomedical data-sharing consent conditions, and its terms are widely adopted in genomics and biobanking to enable semantic interoperability [6,16]. A DUO-based policy combines permissions and modifiers, such as general research use, disease-specific research, or a requirement that ethicsapproval be obtained, into the condition under which access is granted.

DUO Term Categories

Although DUO offers a set of permissions and modifiers, its terms are heterogeneous in how they must be evaluated for compliance. To structure their handling in a zero-knowledge setting, we group DUO terms into four categories:

- **Boolean-only:** simple yes/no conditions that can be represented directly without external validation, e.g. “no restriction” (NRES), general research use (GRU), or “health or medical or biomedical research” (HMB).
- **Ontology-linked:** terms whose evaluation involves consistency checks against external vocabularies or classifications, e.g. disease-specific research (DS), population origins or ancestry research only (POA), or clinical care use (CC).
- **Obligation-required:** terms that impose procedural or temporal obligations, e.g. “publication required” (PUB), “publication moratorium” (MOR), “time limit on use” (TS), or “return to database or resource” (RTN).
- **Credential-required:** terms whose evaluation relies on attested proof of requester status or approvals, e.g. “collaboration required” (COL), “ethics approval required” (IRB), “project-specific restriction” (PS), or “institution-specific restriction” (IS).

Table 1 summarizes representative DUO terms under this category. Some terms reduce to simple Boolean flags, while others rely on ontology references, external obligations, or credential attestations.

3. Related Work

Blockchain-based consent management for healthcare data sharing has been studied extensively, with smart contracts used to record, enforce, and audit consent across institutions [1–3]. These works focus on making consent decisions verifiable and auditable, and the privacy mechanisms surveyed in this domain mainly protect health data contents and user identities [5]. The confidentiality of the governing consent policy, and what public-chain enforcement artifacts reveal about it, is typically out of scope.

Beyond consent management, related work addresses individual requirements of policy-private consent enforcement (policy confidentiality, metadata-leakage control, ver-

Table 1. Representative DUO terms grouped by evaluation category.

Category	Example Terms	Description
Boolean-only	NRES (DUO:0000004), GRU (DUO:0000042), HMB (DUO:0000006)	Direct yes/no conditions, no external validation required.
Ontology-linked	DS (DUO:0000007), POA (DUO:0000011), CC (DUO:0000043)	Evaluated against external ontologies such as the International Classification of Diseases (ICD-10) or the Mondo Disease Ontology (MONDO).
Obligation-required	PUB (DUO:0000019), MOR (DUO:0000024), TS (DUO:0000025), RTN (DUO:0000029)	Impose obligations such as publication duties, time limits, or data return.
Credential-required	COL (DUO:0000020), IRB (DUO:0000021), PS (DUO:0000027), IS (DUO:0000028)	Depend on credentials, approvals, or institutional/project-specific restrictions.

ifiable auditability, or a stable verification interface), but none of the approaches reviewed here combines them on a public chain. Encrypted data-sharing mechanisms, such as CP-ABE and PRE [17,18], control access to encrypted data. However, as noted in Section 2.1, standard CP-ABE exposes the policy itself, while PRE-based delegation may reveal authorization patterns that data encryption is not designed to hide. Transaction-privacy systems, including Zerocash-style protocols [22] and Ethereum-oriented systems such as ZETH and AZTEC [7,8], provide public verification over private transaction witnesses, but they mainly target asset ownership, values, or participant relationships rather than compliance with a provider-defined consent policy.

Hidden-policy CP-ABE schemes reduce direct policy exposure by hiding the access structure in the ciphertext [13,14]. Wu et al. [23] instead realize attribute-based access control (ABAC) by encrypting policy and attribute values under an additively homomorphic cryptosystem. In this scheme, a smart contract evaluates access conditions over encrypted values with assistance from each organization's node, which provides partial decryption results and zero-knowledge proofs of their correctness. However, the Boolean structure of the policy and its attribute names remain visible on the ledger, and per-attribute match results can be recovered from the ciphertexts and partial decryptions recorded with each decision [23]. The scheme also operates on a consortium blockchain, where enforcement artifacts are not globally visible as they are on a public blockchain. Other approaches provide broader forms of confidentiality. Functional encryption reveals function outputs while hiding inputs [24], but does not by itself provide publicly verifiable authorization evidence. Private-state blockchain systems, including Secret Network and Oasis/Ekiden, protect smart-contract state or execution through encrypted state and trusted execution environments (TEEs) [25,26]. Policy or state confidentiality alone, however, does not ensure that public-chain enforcement artifacts reveal no additional policy information. TPS-ZKPI therefore focuses on limiting policy information exposed during public-chain consent enforcement while preserving public verifiability of each authorization decision.

Self-sovereign identity (SSI) architectures give holders control over their verifiable claims, including the ability to determine which verifiers may access them. Buccafurri et al. [27] support this control through ciphertext-policy attribute-based proxy re-encryption. The holder stores encrypted claims off-chain under a self-defined attribute policy, and a

semi-trusted, non-colluding consensus provider re-encrypts them for verifiers whose certified attributes satisfy that policy. A smart contract notarizes hashes of storage, policy-change, and access operations so that they can later be attributed to their parties, and a trusted key-generation authority issues all attribute keys. Zero-knowledge credential presentations likewise let a patient authenticate anonymously on public or private chains [28], and surveys of zero-knowledge proofs for blockchain privacy catalogue this design space [12,29]. TPS-ZKPI focuses on protecting the provider's consent policy during authorization. The requester's claims serve as private witness inputs to a reusable zero-knowledge circuit that verifies satisfaction of the committed policy.

Privacy-preserving access-control and policy-evaluation mechanisms have been studied in credential-based settings built on the eXtensible Access Control Markup Language (XACML) and the Security Assertion Markup Language (SAML) [30], as well as in policy evaluation by multi-party computation (MPC) using homomorphic encryption (HE) and secure function evaluation [31]. These approaches support confidential authorization computation, but they are not primarily designed for settings where policy-evaluation artifacts (calldata, public inputs, emitted events, timing, and gas usage) are permanently and globally visible on a public blockchain. In addition, these approaches do not directly provide compact evidence that a smart contract can verify against a committed, provider-defined policy instance.

Reusable proving systems, including universal-setup succinct non-interactive arguments of knowledge (SNARKs), composable proof systems, and zero-knowledge virtual machines (zkVMs) [32–34], improve maintainability by reducing the need for policy-specific circuits or verifier contracts. However, reusable proving alone does not guarantee policy privacy during public-chain consent enforcement. Policy-dependent program representations, public inputs, calldata layouts, logs, or gas profiles may still reveal information about the hidden policy. This motivates TPS-ZKPI's use of a domain-specific reusable policy interpreter, which preserves proof reuse while exposing only a policy-independent public verification interface.

Studies of metadata leakage on public chains show that cryptographic privacy guarantees may be weakened by observable transaction-level artifacts such as timing and volume patterns [4]. Techniques such as batching, decoys, and traffic shaping can reduce some observable signals [35], but they are usually designed for payment privacy, anonymity sets, or transaction-linkability settings. They do not directly provide hidden-policy compliance verification or auditable consent enforcement. In contrast, TPS-ZKPI treats metadata leakage as part of the policy-privacy objective rather than as a separate transaction-level concern.

Table 2 summarizes the reviewed approaches by policy confidentiality, metadata-leakage mitigation, and verifiable auditability, and notes the scope and limitations of each. Partial indicates limited support or a need for application-specific design; N/A denotes a criterion that does not apply to the cited approach.

4. Problem Formulation

4.1. System and Trust Model

We consider a blockchain-based healthcare data-sharing system with two user roles, following the consent model of Jaiman and Urovi [37]: *data providers*, who provide protected resources and govern access through consent policies, and *data requesters*, who seek access to those resources subject to the applicable policies. TPS-ZKPI mediates this interaction through three components: a TPS operating off-chain, an on-chain access-control manager (ACM), and a data access layer. The provider, which may be the data subject or an organization holding the data [37], defines a consent policy P for a protected resource

Table 2. Positioning of representative approaches against policy-private consent enforcement under public-chain observability.

Approach	Policy confidentiality	Metadata-leakage mitigation	Verifiable auditability	Scope and limitations
Blockchain consent management [1,2]	No	N/A	Yes	Records and enforces consent verifiably on chain, but policy confidentiality and policy-dependent leakage are typically out of scope.
Data / transaction privacy [7,8,17,18,22,36]	No	Partial	Partial	Protects encrypted data access or transaction semantics, but not the governing consent policy or policy-dependent authorization workflow.
Hidden-policy ABE [13,14]	Partial	Partial	Partial	Reduces direct policy exposure, but may leak through workflows, updates, or policy-holder interactions.
Homomorphic hidden-policy ABAC [23]	Partial	N/A	Yes	Hides policy and attribute values on a permissioned ledger, but the Boolean structure, attribute names, and per-attribute match results remain recoverable from it.
SSI verifier authorization [27]	No	Partial	Partial	Releases the holder's claims only to attribute-qualified verifiers and notarizes access for later attribution, but protects the requester's claims rather than the provider's consent policy.
MPC / HE policy evaluation [31]	Yes	N/A	N/A	Supports confidential policy evaluation off-chain; the cited design does not specify public-chain enforcement or authorization auditing.
Universal SNARKs / zkVMs [32–34]	Partial	Partial	Yes	Supports reusable proving and public verification; policy hiding and a fixed verification interface require application-specific design.
Metadata-leakage defenses [35]	N/A	Yes	Partial	Mitigates transaction timing and volume leakage and supports verifiable execution, but does not directly provide hidden-policy consent enforcement.
TPS-ZKPI	Yes (under TPS trust)	Yes (modeled channels)	Yes	Supports bounded Boolean policies; confidentiality requires trust in the TPS.

and registers an on-chain commitment h to the encoded policy. The requester seeks access by submitting policy-relevant authorization evidence, such as attributes, credentials, and access context. The TPS constructs the private witness and generates a zero-knowledge proof of policy compliance. The ACM verifies the proof on-chain and records the authorization outcome, while the data access layer grants access according to the recorded decision.

TPS-ZKPI relies on the following assumptions. First, the provider defines the consent policy and commits to the corresponding policy instance on-chain. Second, the TPS is trusted to preserve the confidentiality of policy-bound proving material and requester-submitted authorization evidence, and to follow the prescribed proof-generation workflow. Third, the commitment scheme is assumed to be hiding and binding, and the zero-knowledge proof system is assumed to satisfy completeness, soundness, and zero knowledge with respect to the implemented policy-compliance relation. Issuer governance, credential issuance and revocation, and real-world identity verification are treated as external lifecycle functions.

Conceptually, the TPS is an independent proof service, distinct from the provider; this separation is what makes relocating trust meaningful, since it prevents the provider from being the single party that holds both the policy and the requester evidence. Possible instantiations of such an independent service include a third-party proof service, a TEE-backed proof service that makes the trust assumption hardware-attested, or MPC/-

collaborative SNARK proof generation in which no single party sees both policy material and requester evidence. For evaluation simplicity, the current evaluation co-locates the TPS with the provider (a provider-operated TPS). This co-location concentrates trust in the provider, which holds both the policy and the requester's authorization evidence. A compromised provider-operated TPS could substitute inputs if their origin is not independently authenticated, or refuse to generate proofs for legitimate requesters.

The policy commitment and on-chain proof verification ensure that accepted proofs demonstrate satisfaction of the committed policy relation, regardless of how the TPS is instantiated. However, they do not guarantee the provenance of evidence that is not issuer-signed, which remains dependent on the TPS (Table 3). Policy confidentiality, in contrast, is not cryptographically enforced against the TPS itself; it rests on the trust assumption stated above.

Table 3 summarizes, for each actor or failure mode, how each of four security properties (policy confidentiality, authorization correctness, liveness, and decision auditability) holds: whether it is enforced cryptographically, supported only empirically, dependent on the TPS trust assumption, left to external functions, or not provided by the prototype.

Attribute-origin guarantees depend on the evidence type. For DUO terms evaluated from the claims in an issuer-signed profile credential presented by the requester (hereafter, profile-backed terms), the prototype verifies the credential's signature under an issuer key selected by the provider and bound to the policy commitment. The signature scheme, EdDSA-Poseidon, is the Edwards-curve Digital Signature Algorithm (EdDSA) [38] over the Baby Jubjub curve [39], instantiated with Poseidon [40], a hash function designed for efficient evaluation inside zero-knowledge circuits. The credential binds the policy commitment, requester pseudonym, protected resource, claims, validity interval, and credential identifier; the corresponding policy attributes are derived inside the circuit. Thus, subject to signature unforgeability and Poseidon collision resistance, these values cannot be modified without invalidating the proof.

4.2. Public-Chain Adversary

The primary adversary is a passive public-chain observer. The adversary can inspect smart-contract code and monitor public blockchain artifacts across multiple authorization sessions, including policy commitments, submitted proofs, public inputs, emitted events, calldata, timing, and gas usage. The adversary does not observe provider-to-TPS policy material, commitment-opening randomness, requester-submitted authorization evidence, credential witnesses, or the TPS's witness construction process. These data and computations remain off-chain and are not exposed in the public-chain transcript. We assume the off-chain channels themselves are confidential (e.g., TLS-protected).

Each policy-enforcement instance corresponds to an authorization session, in which a requester seeks access to a protected resource under a committed policy. For each such session, the adversary observes a public view

$$view_{\text{pub}} = (h, \pi, o, pp, d_\chi, \tau, id_{\text{req}}, id_{\text{res}}, \nu, meta), \tag{1}$$

where h is the on-chain commitment to the hidden policy, π is the zero-knowledge proof submitted for on-chain verification, o, pp, d_χ, and τ are the remaining public signals of the proof statement (the authorization output, a parameter commitment, a context digest, and a request epoch; Section 5), id_{req}, id_{res}, and ν are the requester, resource, and session-nonce values carried in the access transaction, and *meta* denotes transaction-level metadata exposed by the enforcement layer, such as calldata footprint, event structure, public-input layout, timing, and gas usage. The adversary's goal is to infer hidden information about the committed policy or the authorization interaction from this public view.

Table 3. Security properties of the TPS-ZKPI prototype against each actor or failure mode, and what each rests on: cryptography, empirical evaluation, the TPS following its workflow, or external functions.

Actor or failure mode	Policy confidentiality	Authorization correctness	Liveness	Decision auditability
Passive public-chain observer	$C_{zk,hid}$ for proof and commitment contents; empirical for modeled metadata	$C_{snd,bind}$	N/A	C_{chain}: outcome recorded
Untrusted requester	As for the observer; adaptive-query leakage not evaluated	$C_{snd,ctx}$: cannot forge a proof or alter its bound authorization context	N/A	C_{chain}: pseudonymous request context and outcome recorded on-chain
Workflow-following TPS	✗ against the TPS itself; T against onward disclosure	$C_{snd,bind}$ for the committed relation; T/E for unsigned evidence provenance	T: proves and submits every request	C_{chain} for submitted decisions; ✗ for refusal
Compromised or colluding TPS	✗	$C_{snd,bind,sig}$ for the committed relation and signed claims; E for externally authenticated evidence; ✗ for unauthenticated evidence provenance	✗ if the TPS refuses to prove	C_{chain} for submitted decisions; ✗ for refusal
Forged credential	N/A	$C_{snd,bind,sig}$: signature verified under the policy-bound issuer key	N/A	N/A
Unauthorized or revoked issuer	N/A	E: issuer authorization and revocation status are external	N/A	C_{chain} for the on-chain decision record; E for determining issuer status

C: cryptographic, with subscripts naming the mechanisms it rests on: zk and snd, zero knowledge and soundness of Groth16; hid and bind, hiding and binding of the policy commitment h and the parameter commitment pp; sig, EdDSA-Poseidon signature unforgeability; ctx, on-chain recomputation of the context digest d_χ; chain, integrity of the on-chain record, assuming the underlying chain is secure. T: trust in the TPS following the prescribed workflow; E: external or lifecycle function; ✗: not provided by the prototype; N/A: not applicable. In the evaluated provider-operated deployment, the provider and TPS are co-located, so all TPS-dependent confidentiality and liveness assumptions are concentrated at the provider.

In addition to the public-chain observer, we consider the requester as untrusted with respect to policy confidentiality. The requester initiates access requests, observes its own submitted evidence and authorization outcome, and inspects the public-chain transcript. However, the requester does not receive the encoded policy, commitment-opening randomness, or policy-bound proving material, which remain within the provider-to-TPS proof-generation boundary. This restriction matters because a requester that learned the policy could infer sensitive facts about the data subject from the consent terms themselves and, in addition, would learn the policy's full acceptance region (which attribute combinations are granted), enabling targeted evidence construction or probing beyond its own legitimate sessions.

4.3. Privacy Objectives

We operationalize consent policy privacy as three measurable, attack-oriented objectives that cover the information a public-chain observer could extract about a hidden pol-

icy: its semantic content, which of several similar policies it is, and its structural form. These correspond to Policy Content Hiding, Policy Indistinguishability, and Structure and Parameter Privacy, respectively.

Policy Content Hiding (PCH). PCH requires that public-chain artifacts should not reveal the semantic content of the hidden consent policy, such as data-use terms, disease categories, research purposes, institutional restrictions, or ethics-related requirements.

Policy Indistinguishability (PI). PI requires that different candidate policies should not become distinguishable from their public verification artifacts when they are compiled into the same fixed public interface. This captures whether two policies can be told apart from commitments, proofs, public parameters, or metadata even when their direct representations are hidden, that is, whether an observer can tell which policy governs an execution.

Structure and Parameter Privacy (SPP). SPP requires that public-chain artifacts should not reveal structural or parametric properties of the hidden policy, such as logical structure, nesting depth, predicate count, or set size.

4.4. Problem Statement

Given P, requester authorization evidence, a globally observable public blockchain, and the TPS trust assumption stated in Section 4.1, the problem is to enforce access to a protected healthcare resource subject to four requirements. First, authorization must be publicly verifiable. The ACM and any observer must be able to verify that the requester satisfies the committed policy without learning the policy or executing it in plaintext. An accepted proof must be bound to the committed policy and the request's authorization context, including the requester pseudonym, resource, and session, to prevent reuse in a different context. Second, the policy must remain confidential. Public-chain artifacts should protect policy content, structure, and parameters and preserve policy indistinguishability as defined by PCH, SPP, and PI (Section 4.3). Requesters should learn no policy information beyond what their own authorization outcomes reveal, and requester evidence should remain off-chain. Third, authorization decisions should be auditable. Each processed request, whether granted or denied, should be recorded on-chain together with its bound authorization context, allowing the decision to be verified later without revealing the policy logic. Fourth, one policy-independent verification interface should serve every policy in the supported class, so that policies can be added or changed without changing the on-chain verifier.

5. The Proposed Approach: TPS-ZKPI

5.1. Overview

As shown in Figure 1, the TPS-ZKPI workflow binds a provider-committed hidden policy to requester-submitted authorization evidence through TPS-side proof generation and on-chain proof verification. The provider registers only a policy commitment on-chain and sends policy-bound proving material to the TPS off-chain (steps 1–2 in the figure). For each access request, the TPS constructs the private witness from the hidden policy material and requester evidence, generates a zero-knowledge proof of the access decision, and submits a fixed-format proof statement for ACM verification (steps 3–4). The ACM records the authorization outcome in contract storage and emits it in the blockchain event log (as access-granted or access-denied events; step 5); the data access layer consults the recorded decision to serve or refuse the request (steps 6–9).

This separates confidentiality-sensitive proof generation from public verification. Policy material and requester evidence remain off-chain (policy material within the provider–TPS boundary, requester evidence between the requester and the TPS), while the

blockchain verifies only a fixed-format proof statement against the provider's policy commitment, alongside session parameters validated by the ACM. In a traditional requester-as-prover design, the requester generates the proof itself and must therefore receive the encoded policy and the commitment-opening randomness, so every requester learns the policy in normal operation. TPS-ZKPI keeps this material within the provider–TPS boundary, so requesters do not learn the policy (Section 4.1).

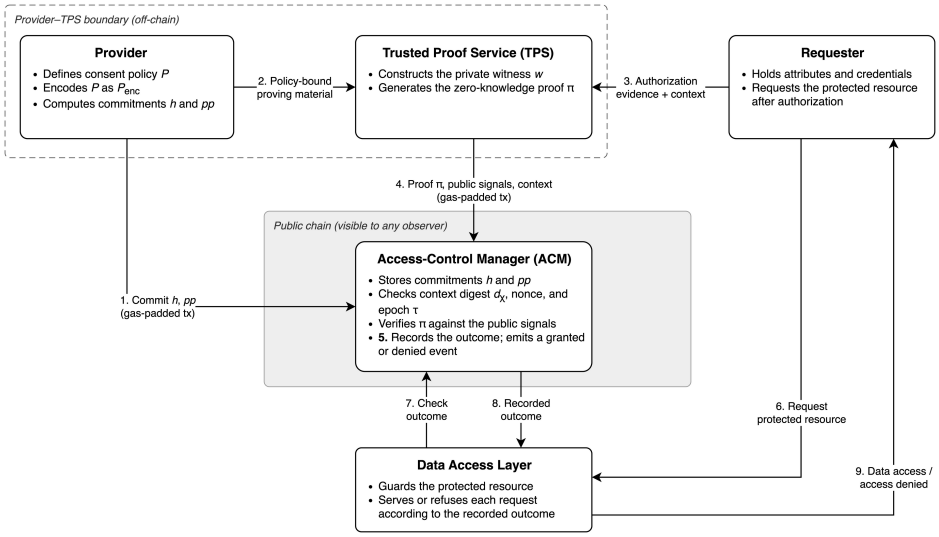


Figure 1. An overview of the TPS-ZKPI workflow.

5.2. Policy Representation and Interpretation

TPS-ZKPI represents heterogeneous Boolean consent policies in a fixed-length, circuit-friendly format. This policy class is chosen for two reasons. First, the supported biomedical consent conditions (DUO-style data-use terms, disease-scope constraints, and requester restrictions) are categorical conditions combined with AND/OR, and conditions that are not natively Boolean enter through fixed external-predicate slots. The generic interpreter also evaluates NOT, but the DUO compiler does not expose negation because its membership witnesses establish inclusion rather than non-membership. Second, a Boolean policy class admits a fixed-length encoding evaluated by a single reusable interpreter circuit, which keeps the verification interface policy-independent, a prerequisite for the privacy objectives evaluated later. Let

$$X = \{0,1\}^N, \qquad B = \{0,1\}^M, \tag{2}$$

denote the requester-attribute assignment space and the external-predicate assignment space, respectively. The assignment space is

$$O = X \times B, \tag{3}$$

where $a \in X$ encodes requester attributes and $b \in B$ encodes predicate outcomes, including credential, ontology-membership, requester-allowlist, and obligation-related checks. The implemented relation derives these bits from their associated witnesses rather than accepting unconstrained predicate assertions.

Ontology-linked conditions are represented extensionally rather than through in-circuit ontology reasoning. At encoding time, the provider expands an ontology concept into a whitelist of codes and commits the whitelist as a Merkle root among the policy's per-predicate parameters; in-circuit, the predicate check is a set-membership proof of the

requested term against this root. A disease-specific policy scoped to diabetes, for example, whitelists the diabetes subtree of the International Classification of Diseases, 10th Revision (ICD-10), whereas one scoped to type-1 diabetes whitelists only the type-1 codes. Expanding the concept hierarchy at encoding time keeps ontology reasoning out of the circuit and preserves the policy-independent interface.

Consent policies are expressed as Boolean formulas over assignment bits:

$$P ::= atom \mid (P \wedge P) \mid (P \vee P) \mid \neg P, \qquad atom ::= x_i \mid y_j. \tag{4}$$

Here x_i ($1 \leq i \leq N$) selects requester-attribute bit a_i and y_j ($1 \leq j \leq M$) selects external-predicate bit b_j. For example, the consent condition "the resource may be used for disease-specific research on diabetes, or for clinical care under approved clinical terms" is expressed as $(x_1 \wedge y_1) \vee (x_2 \wedge y_2)$, where x_1 and x_2 flag the disease-specific-research and clinical-care purposes among the requester attributes, y_1 attests that the requested disease code lies in the committed diabetes whitelist, and y_2 attests that the requested clinical term lies in the approved clinical whitelist. Each policy is compiled into an RPN program over operand tokens and Boolean operator tokens. RPN linearizes arbitrarily nested formulas into a parenthesis-free token sequence that a stack machine evaluates in a single left-to-right pass with a fixed step count, a control structure that maps directly onto a fixed-size circuit and lets one interpreter serve all policies. To support reusable verification, each program is padded to a fixed token length L:

$$P_{\text{enc}} = [t_1, \ldots, t_L], \tag{5}$$

where $t_1, \ldots, t_\ell$ are operand tokens (x_i or y_j) or operator tokens ($\wedge$, $\vee$, $\neg$), together with an explicit program length $\ell \leq L$ carried as a private witness. The interpreter evaluates only the first ℓ tokens; padding positions beyond ℓ are inactive and leave the output unchanged. The example policy above compiles to the token sequence $[x_1, y_1, \wedge, x_2, y_2, \wedge, \vee]$ with $\ell = 7$; the remaining $L - 7$ positions are padding. The implemented circuit uses $N = 13$, $M = 11$, and $L = 64$. We denote the set of length-L programs by Prog_L.

The reusable policy interpreter is a zero-knowledge circuit

$$C : \mathsf{Prog}_L \times \{0, \ldots, L\} \times O \to \{0, 1\}. \tag{6}$$

Given an encoded policy, its program length, and operand values, the interpreter evaluates the RPN program using deterministic stack semantics. The circuit constraints enforce program well-formedness, so no satisfying witness exists for a program that violates the stack discipline. For a well-formed program, the circuit outputs 1 if the requester's evidence satisfies the encoded policy and 0 otherwise. The acceptance relation requires that C output 1.

A new or changed policy is therefore only a different program for the same circuit, so policies can evolve without changing the on-chain verifier.

Scope of the supported policies. Table 4 summarizes how the prototype supports each consent feature (in-circuit, on-chain, committed only, or not provided) and which combinations are supported. In particular, the ACM enforces time-limited use (TS) using public state. The commit transaction emits the duration, and the first grant emits the requester's authorization window, so policies containing TS do not provide content privacy for that term.

Table 4. Consent-policy features and how far the prototype establishes them.

Consent feature	Enforcement	What is established at authorization time
Unrestricted use (NRES)	In-circuit	Compiled as a standalone unconditional policy; it cannot be combined with another DUO term.
Categorical data-use terms combined with AND/OR	In-circuit	The committed program and its length are evaluated by the interpreter; each term may appear at most once.
Disease, study-design, ancestry, geography, and clinical-term scopes (DS, RS, POA, GS, CC)	In-circuit	Merkle membership of the requested code in a provider-committed whitelist whose root is bound in the parameter commitment.
User-, project-, and institution-specific restrictions (US, PS, IS)	In-circuit	Membership of the holder pseudonym in a provider-committed allowlist for the scope.
Profile-backed terms (GRU, HMB, NMDS, GSO, NPUNCU, NPOA, NPU, NCU; Section 5.4)	In-circuit	EdDSA-Poseidon credential over policy commitment, pseudonym, resource, claims, validity interval, and identifier; the attributes are derived from the signed claims, but subsequent conformity with a declared purpose is not established.
Policy validity window	In-circuit + on-chain	The committed window is checked against the public request epoch, which the ACM requires to equal the current hour.
Time-limited use (TS)	On-chain	The committed duration is also stored and emitted in clear by the ACM; it starts a per-requester window on the first grant, and later requests outside it are denied. TS may appear only conjunctively.
Publication moratorium (MOR)	Committed	The deadline is bound into the policy as an accepted obligation; publication conduct is not observable at request time. MOR may appear only conjunctively.
Prospective consent withdrawal and policy evolution	On-chain	A successor policy supersedes its predecessor, a policy can be revoked, and a resource can be rebound for future decisions; this does not retract data already released or enforce downstream cessation or deletion.
Institutional or ethics approval (IRB)	Not provided	Documentary approval and committee review remain an out-of-circuit authorization step.
Downstream purpose limitation, post-access duties (PUB, COL, RTN), delegation, credential revocation	Not provided	Carried by external governance, the data access agreement, and the issuer lifecycle, outside the proved statement.

"In-circuit" means the condition is part of the proved relation; "on-chain" means the ACM enforces it from public state; "committed" means the value is bound into the policy but compliance is not decided at request time.

5.3. Commitment Binding and Proof Generation

To bind each proof to a provider-defined hidden policy, the provider samples commitment randomness r uniformly from the scalar field of BN254 [41,42], the pairing-friendly elliptic curve over which the proofs are computed, and computes

$$\begin{aligned} q &= \mathsf{H}_{\mathrm{pol}}(P_{\mathrm{enc}}), \\ h &= \mathsf{Com}(P_{\mathrm{enc}}, \ell; r) = \mathsf{Poseidon}(q \parallel \ell \parallel r), \end{aligned} \tag{7}$$

where $\parallel$ denotes concatenation of field-element inputs, q is the Poseidon-based digest of the fixed-length encoded policy, ℓ is the private program length, r is the commitment randomness, and h is the policy commitment stored on-chain[2]. The semantics of an encoded policy are determined by the pair (P_{enc}, ℓ), since ℓ identifies the active prefix of the padded token array. Accordingly, the circuit supplies the same private value ℓ to both the commitment verifier and the policy interpreter. Assuming Poseidon is collision-resistant, binding the proof to h prevents the prover from changing the committed program length. This pre-

[2] For example, the commitment $h =$ `0x13bd...e781` of the first policy in the evaluation dataset is recorded in the `PolicyCommitted` event of the Amoy transaction https://amoy.polygonscan.com/tx/0x7b2e8238c0edb0aca858dbd4e1dca0ce0795e1fe3ae03ff6670abec697f60d6a#eventlog.

vents the prover from truncating P_{enc} to a shorter prefix that satisfies the stack constraints but skips part of the policy evaluation. This argument relies on the fixed-length encoding of P_{enc}, which keeps the sponge input unambiguous. Hiding is an explicit commitment assumption (Section 4.1) that relies on the high-entropy randomness r.

For an access request, let

$$\chi = (id_{\text{req}}, id_{\text{res}}, \nu), \qquad d_\chi = \mathsf{Poseidon}(id_{\text{req}} \parallel id_{\text{res}} \parallel \nu), \tag{8}$$

where ν is a fresh session nonce. Here id_{req} is the requester pseudonym. For policies that activate a holder-bound credential or requester-allowlist predicate, the circuit derives it as $\mathsf{Poseidon}(s_{\text{holder}})$ from a private holder secret s_{holder}; otherwise, it may be a requester-supplied pseudonym. The TPS constructs the private circuit witness

$$w = (P_{\text{enc}}, \ell, a, b, r, \omega, \chi), \tag{9}$$

where P_{enc} is the encoded policy, ℓ is its program length, a denotes requester attributes, b denotes external predicate outcomes, each constrained to the result of its check over ω, r is the commitment randomness, and ω denotes auxiliary witnesses for credential- or predicate-dependent checks. The context tuple χ consists of private circuit inputs, while its digest d_χ is public. The public-signal vector is

$$u = (o, h, pp, d_\chi, \tau), \tag{10}$$

where o is the authorization output of the circuit, pp is a fixed-format commitment to the per-predicate parameters, and τ is the request epoch used by the temporal checks. Although the three components of χ are not separate Groth16 public inputs, the ACM receives them as transaction parameters in order to recompute d_χ; they are therefore visible in transaction calldata.

For the compiled circuit, the Groth16 [21] setup produces the common reference string as a proving key and a verification key, once for the shared circuit:

$$(pk, vk) \leftarrow \mathsf{Setup}(C). \tag{11}$$

The TPS generates a proof

$$\pi \leftarrow \mathsf{Prove}(pk, u, w). \tag{12}$$

An accepting proof ($o = 1$) certifies

$$\begin{aligned} C(P_{\text{enc}}, \ell, (a, b)) = 1, \qquad h = \mathsf{Com}(P_{\text{enc}}, \ell; r), \\ d_\chi = \mathsf{Poseidon}(id_{\text{req}} \parallel id_{\text{res}} \parallel \nu), \end{aligned} \tag{13}$$

and that the selected external-predicate witnesses satisfy their policy-bound verification relations. Before verifying π, the ACM checks that the resource is governed by the stated policy, that τ is current, and that the nonce has not been used for the same requester–resource context. It also recomputes the digest from the transaction parameters and checks that it matches the public signal d_χ. These checks bind the proof to the requester pseudonym, resource, and nonce within a single ACM deployment, preventing its acceptance under a different combination of these values. The ACM then verifies π against vk and records the outcome of each request. A request is granted only if the proof is valid and $o = 1$; otherwise, it is recorded as a denial.

5.4. Credential-Origin Binding

The circuit reserves one fixed predicate slot for a credential carrying a fixed set of DUO-relevant claims. Let $c_0, \ldots, c_6$ be the private, issuer-signed claim bits for, respectively, research use, health or medical use, methods development, genetic study, ancestry, commercial use, and non-profit use. Let t_{from} and t_{until} define its validity interval, and let γ be its credential identifier. Let d_{DUO} be the fixed credential schema/domain tag for this credential type. From these signed claims, the circuit derives the DUO terms GRU, HMB, no general methods research (NMDS), genetic studies only (GSO), not-for-profit, non-commercial use only (NPUNCU), population origins or ancestry research prohibited (NPOA), not-for-profit organization use only (NPU), and non-commercial use only (NCU). For issuer public key $A = (A_x, A_y)$, the circuit reconstructs the domain-separated message, denoted as:

$$m_{\text{cred}} = \mathsf{Poseidon}(d_{\text{DUO}} \parallel h \parallel id_{\text{req}} \parallel id_{\text{res}} \parallel c_0 \parallel \cdots \parallel c_6 \parallel t_{\text{from}} \parallel t_{\text{until}} \parallel \gamma). \tag{14}$$

Before proof generation, an issuer holding the corresponding private signing key s_{iss} issues the credential by computing

$$\sigma \leftarrow \mathsf{Sign}_{\text{EdDSA-Poseidon}}(s_{\text{iss}}, m_{\text{cred}}). \tag{15}$$

The resulting credential is supplied to the TPS as private off-chain evidence for proof generation. At proof generation, the TPS supplies A, σ, and the credential fields as private witness values. The circuit then enforces

$$\mathsf{Verify}_{\text{EdDSA-Poseidon}}(A, \sigma, m_{\text{cred}}) = 1, \qquad \mathsf{Poseidon}(A_x \parallel A_y) = k_{\text{iss}}, \tag{16}$$

where k_{iss} is the issuer-key commitment contained in the policy-bound parameter commitment pp. The circuit additionally requires $t_{\text{from}} \leq \tau \leq t_{\text{until}}$ and $\gamma \neq 0$. Because this credential is holder-bound, a policy that activates this slot also requires $id_{\text{req}} = \mathsf{Poseidon}(s_{\text{holder}})$, binding the credential to the requester pseudonym in the authorization context. The corresponding DUO attributes are constrained to these claim-derived values.

5.5. Transcript Normalization

Zero-knowledge proofs hide private witness values, but surrounding blockchain artifacts may still leak policy-dependent information to the passive public-chain observer (Section 4). TPS-ZKPI therefore normalizes the public-chain transcript so that different policies induce fixed or tightly bounded observable profiles. Table 5 summarizes the mechanisms.

Table 5. Transcript-normalization mechanisms in TPS-ZKPI.

Artifact	Normalization mechanism
Calldata	Fixed verification interface and fixed-size proof/public-input encoding.
Logs	Fixed event structure with constant topic and payload layout.
Public parameters	Fixed-format handles instead of policy artifacts or credential witnesses.
Gas usage	Policy-independent verifier logic and gas padding that counts calldata byte costs and tops each transaction up to a target drawn at random from a fixed band.

6. Experiments and Results

We evaluate PCH, PI, and SPP (Section 4.3) against a passive public-chain observer (Section 6.1). To assess whether this observer can infer hidden policy information from the on-chain observables, we combine supervised attacks with an unsupervised cluster-

ing cross-check. The supervised attacks attempt to predict the hidden labels, while the unsupervised analysis groups executions based on their observables without using those labels. We then assess whether the resulting clusters correspond to the hidden labels.

Table 6 summarizes the supervised results across the three privacy objectives. Under the leakage criterion, 43 of the 45 supervised tasks are classified as PASS and the remaining two as FAIL. The unsupervised analysis shows little agreement between the clusters and the hidden labels, with only a weak clustering signal observed. Diagnostic analyses suggest that the two FAIL outcomes and this weak signal reflect chance associations between the randomly drawn padding targets and the hidden labels.

Beyond privacy, we evaluate the system's computational and on-chain costs, including gas consumption for commit and access transactions, padding overhead, calldata sizes, circuit constraint count, and proving time. We also examine how these costs vary with circuit capacity and policy complexity within a fixed circuit, and how offline proof generation throughput scales with workload.

Table 6. Supervised attack results on the Polygon Amoy dataset.

Objective	Hidden target	Passing	Results (passing tasks)
PCH	DUO terms and ICD-10 chapters	19/20	AUC 0.464–0.570, median significant-seed fraction 3%
PI	Which policy of a related pair is executed	3/3	AUC 0.509–0.549, median significant-seed fraction 7%
SPP	Structure, predicate, parameter, and nesting-depth buckets	20/21	AUC 0.468–0.571, median significant-seed fraction 7%

Significant-seed fraction: share of the 30 policy-disjoint splits (seeds) whose permutation test rejects at $\alpha = 0.05$ (Section 6.2). Passing: tasks not flagged by the leakage criterion, which flags a task when its mean AUC exceeds 0.55 and its significant-seed fraction is at least 50%. Failing tasks: PCH ICD-10 ch5 (AUC 0.667, significant-seed fraction 53%) and SPP `nestingDepth medium/deep` (AUC 0.563, 53%); both trace to the padding target (Sections 6.4 and 6.6).

6.1. Adversary and Observables

We consider the passive public-chain observer defined in Section 4. This adversary has unrestricted read access to the ledger and observes all transactions and events generated by TPS-ZKPI, but has no direct access to the plaintext policy or the witness. Under the assumptions stated in Section 4.1, the zero-knowledge proof reveals no information beyond the public statement, and the hiding commitments conceal the committed policy. These cryptographic protections do not, by themselves, prevent policy information from leaking through observable metadata.

We consider three adversary goals, one per privacy objective: (i) recover the policy's *content* (PCH, Section 6.4); (ii) distinguish two *semantically related* policies (PI, Section 6.5); and (iii) recover the policy's *structure or parameter sizes* (SPP, Section 6.6).

To evaluate attacks against these goals, we extract the following metadata features from each policy execution, consisting of a commit transaction followed by an access transaction:

- **Gas:** `commitGasUsed`, `accessGasUsed`.
- **Calldata length:** `commitCalldataLength`, `accessCalldataLength`.
- **Proof size:** `proofSize`.
- **Event logs:** `commitLogsCount`, `commitLogsSize`, `accessLogsCount`, `accessLogsSize`.

The fixed circuit and commitment-only interface keep calldata length, proof size, and event-log counts and sizes constant across the evaluated policies. These features remain

in the attack pipeline but provide no variation for the attacks to exploit. Any policy-dependent variation in the selected feature set must therefore occur in the two gas measurements (`commitGasUsed` and `accessGasUsed`).

6.2. Attack Methodology

We evaluate each privacy objective with a primary supervised attack and a complementary unsupervised cross-check. The *supervised* method trains classifiers on labeled data to predict hidden labels from the observables. Performance is measured by AUC, with a chance baseline of 0.5. The *unsupervised* method clusters the observables without using labels. We evaluate agreement between the resulting clusters and the true labels using the Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI).

We execute the generated policies on the *Polygon Amoy* testnet. For each policy, a provider account submits the policy commitment using `commitPolicyWithGasPad`. We then generate the Groth16 proof locally, acting as the TPS (Section 4.1). A separate TPS account submits the access transaction using `requestAccessBoundWithGasPad`. In the prototype, provider-to-TPS and requester-to-TPS exchanges are implemented as in-process calls and generate no additional on-chain artifacts. We extract the observables defined in Section 6.1 from the transaction receipts, calldata, and event logs.

6.2.1. Supervised Attacks

We evaluate each objective through binary classification *tasks*, in which the attacker predicts a hidden label from the observables. PCH tasks predict DUO-term presence or ICD-10 chapter membership (Section 6.4). PI tasks distinguish between executions of two policies (Section 6.5). SPP tasks distinguish between two categories of a structural or parameter property (Section 6.6).

For every task, we train two classifiers: logistic regression (LR) and random forest (RF), representing linear and non-linear models, respectively. Both classifiers receive all nine features listed in Section 6.1, including the seven that are constant by design. Each task is evaluated over **30 random seeds**, using the same protocol across PCH, PI, and SPP.

For each seed, we first randomly select one execution per policy to limit the influence of repeatedly executed policies on the AUC and its permutation test. We then construct a **policy-disjoint** $70/30$ **train/test split**, ensuring that no policy appears in both sets. Test AUC therefore measures attack performance on policies not seen during training. Because policies are kept intact, exact per-class split proportions cannot always be achieved. For each seed, we generate 512 candidate splits and discard those that lack either class in the training or test set. Among the valid candidates, we select the split that best approximates a 30% test fraction within each class.

For each seed and classifier, we assess test AUC using a label-permutation test. After training, we keep the test-set prediction scores fixed and randomly permute the test labels $R = 500$ times. We recompute AUC for each permutation to obtain its null distribution under no association between scores and labels. The empirical p-value is

$$p = \frac{k+1}{R+1}, \tag{17}$$

where k counts the permutations whose AUC is at least as large as the observed AUC. The $+1$ correction prevents a zero p-value, with a minimum attainable value of $1/(R+1)$. The result for a given seed and classifier is considered significant when $p < 0.05$.

All tasks are binary, and the chance baseline for AUC is 0.5 regardless of class balance. Class imbalance may affect estimation uncertainty but does not change this baseline.

For each task, let AUC_s denote the test AUC for seed s, with $S = 30$ seeds. The results report the mean AUC and the AUC gap G_{AUC}, defined as the mean absolute deviation from 0.5:

$$\mathrm{AUC}_{\mathrm{mean}} = \frac{1}{S}\sum_{s=1}^{S}\mathrm{AUC}_s, \qquad G_{\mathrm{AUC}} = \frac{1}{S}\sum_{s=1}^{S}|\mathrm{AUC}_s - 0.5|. \tag{18}$$

Taking absolute deviations before averaging prevents values above and below 0.5 from canceling.

The PASS/FAIL decision uses the mean AUC and the significant-seed fraction. The latter is the fraction of seeds with $p < 0.05$, reported as %sig. We use operational thresholds of 0.55 for mean AUC and 0.5 for the significant-seed fraction. We classify each task as follows:

> **FAIL** if$\mathrm{AUC}_{\mathrm{mean}} > 0.55$ and at least 15 of the 30 seeds yield $p < 0.05$. **PASS** otherwise.

We refer to this rule as the *leakage criterion* and apply it independently to LR and RF. A task is classified as FAIL if either classifier meets both conditions. The result tables report the classifier with the higher mean AUC. Because the splits reuse the same policy dataset, the significant-seed fraction is interpreted as a consistency measure, not a combined p-value.

A PASS means that the evaluated attacks did not meet this criterion for the task. It does not establish the absence of policy leakage.

6.2.2. Unsupervised Clustering Attacks

We run K-Means (with K set to the number of classes, or $K = 2$ for policy pairs) on the on-chain gas features, and score the recovered clusters against the true labels with ARI and NMI. Values near 0 mean the clusters carry no information about the hidden label; values near 1 mean full leakage.

PCH clusters all executions globally, and SPP is clustered the same way on its 1,125-execution view (Section 6.3), once per annotation axis of Section 6.6 with K equal to that axis's bucket count. At these sample sizes, ARI and NMI are read directly, each averaged over 20 K-Means seeds. To test whether an ARI exceeds chance, the cluster assignments are kept fixed, the labels are shuffled 200 times, and p_{ARI} follows Equation (17) with $R = 200$, so its smallest attainable value is $1/201 \approx 0.005$.

For PI, clustering is scored per pair, so each side of a pair must contribute enough executions; this requires two safeguards. First, the PI clustering analysis uses a dedicated set of *balanced policy pairs* constructed from depth-layer executions (Section 6.3). We evaluate 127 pairs, each containing 30 executions per policy. The policy pairs used in the supervised PI attack (Section 6.5) are unsuitable for per-pair clustering. Most pairs contain only one execution per policy, while some contain 1 execution for one policy and 30 for the other. These small or highly imbalanced samples make clustering scores unreliable indicators of policy distinguishability. The supervised attack is unaffected, because it pools the executions of each pair type. Second, NMI is *permutation-debiased*. Even after balancing, each side of a pair holds only 30 executions. At that scale NMI is inflated even without signal, whereas ARI is not (Appendix A), so ARI is read directly and NMI must be debiased. The cluster assignments are computed once (K-Means does not see the labels) and the observed NMI is averaged over 20 K-Means seeds; the pair labels are then shuffled 200 times and the NMI recomputed against the fixed assignments on each shuffle, giving the null distribution of NMI under zero true signal. The debiased NMI is the observed NMI minus the mean of this null distribution, and the empirical p-value p_{NMI} follows Equation (17) with $R = 200$, where k counts the shuffles whose NMI meets or exceeds

the observed value. The cross-check flags a pair only if its debiased NMI is positive *and* significant ($p_{\mathrm{NMI}} < 0.05$).

6.3. Dataset and Corpus Generation

All three privacy evaluations draw on a common policy dataset. The dataset holds 1077 distinct policies run as 1773 on-chain executions on *Polygon Amoy*, each execution being one commit transaction followed by one access transaction. The policies are produced by a deterministic generator with a fixed seed (`unified-decoupled-2026`). For reproducibility, the generator uses a deterministic test issuer to construct private credential witnesses for policies that require credential verification.

For each policy, the generator selects a set of DUO data-use terms and a logical form that combines them. Five terms use code whitelists drawn from external vocabularies: DS uses ICD-10, RS uses the Ontology for Biomedical Investigations (OBI) [43], POA uses the Human Ancestry Ontology (HANCESTRO), GS uses ISO 3166 country codes, and CC uses SNOMED Clinical Terms (SNOMED CT). The other five terms (GRU, HMB, NMDS, GSO, NCU) are evaluated using signed claims from the requester's profile credential (Section 5).

Whitelist sizes are sampled once per policy from a fixed set of candidate sizes, independently of the logical form. The recorded whitelist size is the number of codes included in the committed whitelist.

For example, one generated policy combines the DUO terms RS (research-specific use) and DS (disease-specific use). It requires the study design to belong to the OBI *observational and comparative designs* family and the requested record to fall within the ICD-10 *diseases of the circulatory system* chapter. The two conditions are joined by `AND`, so access is granted only when both are satisfied. Each condition is represented by a predicate that checks membership in a whitelist. The RS predicate uses a whitelist of 8 OBI study-design codes, while the DS predicate uses 8 ICD-10 codes drawn from that chapter. Both whitelist sizes fall within the "medium" parameter-size bucket in Table 7.

We execute the policies in two layers. The *coverage layer* executes 1053 distinct policies once each, spanning the content labels and structural buckets to support training and testing on disjoint policy sets. The *depth layer* executes 24 policies 30 times each (720 executions), providing repeated observations for the per-pair PI clustering analysis (Section 6.2). All 1,773 executions produced the expected authorization outcomes. Sampling is configured to provide at least 30 executions in the smallest content-label class. Table 7 summarizes the counts for each content label and structural bucket.

The evaluated content labels cover ten DUO data-use terms and ten ICD-10 chapters (Table 7). Evaluating every DUO term and individual ICD-10 code would require a substantially larger on-chain dataset. We therefore evaluate DUO-term presence and ICD-10 chapter membership rather than individual ICD-10 codes.

6.4. Policy Content Hiding (PCH)

This attack targets the DUO data-use terms of the corpus, namely GRU, DS, RS, POA, GS, CC, NCU, HMB, NMDS, and GSO (Section 6.3), as well as disease codes aggregated to ten ICD-10 chapters (Table 7).

Each DUO term defines a one-vs-rest binary task, on which we run the supervised pipeline stated in Section 6.2. Table 8 reports the per-term results. Mean AUC ranges from 0.477 to 0.540, below the operational threshold of 0.55. The highest significant-seed fraction is 23% for GRU, below the 50% threshold. All DUO tasks are therefore classified as PASS under the leakage criterion.

The same pipeline applied to the ICD-10 chapters (one-vs-rest over the DS policies only, which carry the ICD-10 scope; 127 test policies per seed; Table 9) passes nine chapters

Table 7. Composition of the shared policy corpus

Grouping	Category	Policies	Executions
DUO data-use term (multi-label)			
	GRU	302	447
	DS	422	770
	RS	412	760
	POA	392	624
	GS	382	556
	CC	392	769
	NCU	206	380
	HMB	331	447
	NMDS	237	469
	GSO	236	352
ICD-10 chapter (policies with an ICD-10 scope, i.e., DS policies)			
	ch0	44	131
	ch1	49	49
	ch2	33	62
	ch3	55	55
	ch4	41	70
	ch5	34	63
	ch6	48	135
	ch7	48	77
	ch8	33	62
	ch9	37	66
	Subtotal	422	770
SPP structureType			
	trivial	130	130
	and_only	294	526
	or_only	303	535
	and_or_mixed	350	582
SPP nestingDepth			
	flat	130	130
	shallow	210	413
	medium	335	509
	deep	402	721
SPP predicateCombo			
	single	257	344
	few	645	1080
	many	175	349
SPP paramType			
	none	147	234
	small	232	406
	medium	352	584
	large	346	549
Corpus total		**1077**	**1773**

ICD-10 labels ch0–ch9 index the ten chapters used: A00–B99, C00–D48, D50–D89, E00–E90, F01–F99, G00–G99, I00–I99, J00–J99, K00–K93, and M00–M99.

and fails one. Chapters ch1, ch2, and ch4 sit just above the AUC line (mean AUC 0.558–0.570) but reproduce in at most 23% of seeds, so they are boundary cases. Chapter ch5 (diseases of the nervous system) crosses both thresholds and is flagged by the criterion: mean AUC 0.667 with 53% of seeds significant.

For each padded access transaction, we decompose the recorded gas as

$$g_{\text{acc}} = g_{\text{pad}} + g_{\text{res}}, \tag{19}$$

where g_{acc} is the recorded access gas (`accessGasUsed`), g_{pad} is the padding target that the TPS draws uniformly from the padding band (Section 5), and $g_{\text{res}} = g_{\text{acc}} - g_{\text{pad}}$ is the residual overhead, the gas that the padded transaction consumes beyond its target. On Polygon Amoy, the residual overhead ranges only from 4,661 to 4,750 gas across all 1,773 executions, so almost all variation in the padded access gas comes from the randomized target. The observer sees only the total gas, whereas the capture log also records each

Table 8. Policy Content Hiding: per-term supervised attack over 30 policy-disjoint seeds (one-vs-rest tasks; higher-mean-AUC classifier per task).

Term	AUC (mean $\pm$ std)	AUC gap (mean)	%sig
DUO GRU	0.540 ± 0.026	0.041	23%
DUO DS	0.514 ± 0.029	0.027	3%
DUO RS	0.498 ± 0.028	0.022	0%
DUO POA	0.477 ± 0.024	0.027	0%
DUO GS	0.502 ± 0.027	0.023	0%
DUO CC	0.515 ± 0.029	0.025	3%
DUO NCU	0.528 ± 0.027	0.031	7%
DUO HMB	0.509 ± 0.027	0.024	3%
DUO NMDS	0.516 ± 0.030	0.025	3%
DUO GSO	0.502 ± 0.031	0.026	3%

In each seed, the test set consists of 30% of the 1077 policies (323 policies), each contributing one execution. The positive class of a task consists of the test policies that carry the term; its median size over seeds ranges from 62 (NCU) to 127 (DS).

Table 9. Policy Content Hiding: per-chapter ICD-10 supervised attack over 30 policy-disjoint seeds (higher-mean-AUC classifier per chapter).

Chapter	AUC (mean $\pm$ std)	AUC gap (mean)	%sig
ICD-10 ch0	0.472 ± 0.078	0.064	0%
ICD-10 ch1	0.570 ± 0.066	0.075	23%
ICD-10 ch2	0.565 ± 0.071	0.081	10%
ICD-10 ch3	0.540 ± 0.061	0.061	7%
ICD-10 ch4	0.558 ± 0.061	0.070	13%
ICD-10 ch5	0.667 ± 0.069	0.167	53%
ICD-10 ch6	0.464 ± 0.077	0.065	0%
ICD-10 ch7	0.525 ± 0.064	0.058	3%
ICD-10 ch8	0.478 ± 0.078	0.063	0%
ICD-10 ch9	0.511 ± 0.074	0.062	0%

In each seed, the test set consists of 30% of the 422 DS policies (127 policies), each contributing one execution. The positive class of a chapter consists of the test policies in that chapter; its median size over seeds ranges from 10 to 17.

padding target. We therefore rerun the analysis on g_{res} to determine whether the ch5 separation remains after removing the randomized component.

Across all 20 PCH tasks, the single-feature rank AUC of the padded access gas, computed over all executions without a classifier, coincides with that of the padding target to three decimal places, while the residual AUC lies between 0.47 and 0.52. For ch5, the gas and target AUCs are both 0.601, whereas the residual AUC is 0.472. The 34 ch5 policies happened to draw targets whose mean is 33,000 gas higher than that of the other 388 DS policies, a 3.2σ deviation for a uniform draw of that size. Because each coverage-layer policy is executed once, the 30 evaluation seeds repeatedly partition the same fixed draws, causing this accidental difference to reproduce across seeds.

Applying the evaluation protocol to the residual gives a mean AUC of 0.556 with 10% significant seeds for LR and 0.561 with 17% significant seeds for RF, both yielding PASS. Applying it to the padding target alone reproduces the observed FAIL (0.686, 63%). Moreover, in the separate unpadded ablation, ch5 is already near chance (AUC 0.468 for LR and 0.455 for RF). Thus, for ch5 specifically, we find no detectable policy-dependent gas signal without padding; the FAIL in the padded corpus is instead caused by an accidental finite-corpus correlation between the random padding targets and the ch5 labels. The unpadded ablation reveals substantial leakage for many other policy contrasts, as shown in Section 6.8.

6.5. Policy Indistinguishability (PI)

To evaluate PI, we pair policies using two quantities. The semantic similarity score is a fixed heuristic that returns 1 when two policies share an identical normalized logic string, that is, the same DUO terms joined by the same Boolean form; the string names the vocabulary of each condition but not its value, so two such policies may restrict, for example, different ICD-10 chapters. For all other couples, the score combines the Jaccard overlap [44] of DUO terms with agreement of the ICD-10, OBI, and other scope fields. The parameter fingerprint consists of the whitelist size, Merkle depth, and proof type.

Each candidate policy couple is assigned to one of three types: `semantic_equivalent` (similarity ≥ 0.999 and different fingerprints), `weak_equivalent` (similarity ≥ 0.95 without that fingerprint difference), and `controlled_difference` (similarity < 0.95; the least similar pairs by construction and hence the easiest to separate, which serve as a positive control and are excluded from the PASS/FAIL decision). We sample a fixed budget of 100 pairs per type, or every available pair when fewer exist (71 `weak_equivalent` pairs).

In this corpus, every couple at or above 0.999 shares the logical form, so the `semantic_equivalent` pairs differ in their parameters and possibly in their scope values. 70 of the 71 `weak_equivalent` pairs share the logical form as well, and 30 of these are content-identical policies that differ only in their commitment randomness; despite its name, this type therefore holds the closest pairs of the corpus.

Each pair is an ordered couple (P_0, P_1), and the attacker's task is to decide, from the on-chain features of a single execution, which policy of the pair produced it. Indistinguishability is thus a two-class problem whose ideal outcome is a guess no better than a random guess, corresponding to a classifier AUC of 0.5.

We evaluate the attacker using policy-disjoint train/test splits across 30 random seeds. The pooled task and both pair-type tasks are classified as PASS under the leakage criterion (Table 10). Attack performance remains close to chance, with low significant-seed fractions across repeated splits.

Table 10. Policy Indistinguishability: single-execution supervised attack over 30 policy-disjoint seeds, by policy-pair type (higher-mean-AUC classifier per type).

Pair type	Pairs	AUC (mean $\pm$ std)	AUC gap (mean)	%sig	med n
overall (all types)	271	0.509 ± 0.035	0.027	3%	162
semantic_equivalent	100	0.549 ± 0.064	0.070	7%	60
weak_equivalent	71	0.544 ± 0.079	0.079	10%	42
controlled_difference †	100	0.544 ± 0.064	0.062	13%	60

med n: median test-set size over seeds. † Reported for reference; excluded from the PI PASS/FAIL decision.

6.6. Structure and Parameter Privacy (SPP)

To evaluate SPP, we annotate each policy along four dimensions that capture the structural and parametric properties described in Section 4.3. The four dimensions are logical structure type (`structureType`), predicate count (`predicateCombo`), parameter size (`paramType`), and nesting depth (`nestingDepth`). Each dimension is divided into discrete categories or value ranges. Specifically, `structureType` classifies the logical form joining the DUO terms (Section 6.3) into four categories: `trivial` (a single term with no Boolean connective, e.g. policy logic `"DS"`), `and_only`, `or_only`, and `and_or_mixed`. `predicateCombo` groups policies by predicate-namespace count into `single` (1), `few` (2–3), and `many` ($\geq$ 4); `paramType` groups policies by whitelist size into `none` (no whitelist), `small` (1–4), `medium` (5–16), and `large` ($\geq$ 17); `nestingDepth` groups policies by the maximum nesting depth of the normalized expression into `flat` ($\leq$ 1), `shallow` (2), `medium` (3), and `deep` ($\geq$ 4). In the normalized expression, each term is joined by AND to the predicate

that checks it (e.g. `"DS"` becomes `(DS & P[ICD10])`). The policy counts for each category in the dataset are listed in Table 7.

Each pair of categories within a dimension defines a binary classification task. Using the observables, the attacker predicts which of the two categories the executed policy belongs to.

Table 11 reports the results for all pairwise classification tasks. For most tasks, mean AUCs are close to the chance level of 0.5, ranging from 0.47 to 0.54. The `paramType` tasks `none/small` and `none/large`, and the `nestingDepth` task `flat/medium`, have mean AUCs of 0.571, 0.568, and 0.555, respectively. Although these values exceed the 0.55 threshold, the corresponding significant-seed fractions are 40%, 33%, and 27%, all below the 50% threshold, so these tasks are classified as PASS. Only the `nestingDepth` task `medium/deep` is flagged under the leakage criterion, with a mean AUC of 0.563 and a significant-seed fraction of 53%.

To investigate this result, we apply the padding diagnostic used for the PCH ch5 case (Section 6.4). For the `medium/deep` task, access gas and the padding target have identical single-feature rank AUCs. Replacing access gas with the residual obtained by subtracting the padding target yields PASS for both LR and RF, whereas using the target alone yields FAIL. These findings are consistent with a chance association between the randomly drawn per-policy padding targets and the nesting-depth categories in the evaluated dataset.

Table 11. Structure and Parameter Privacy: 21 pairwise contrasts across four annotation axes over 30 policy-disjoint seeds (higher-mean-AUC classifier per contrast).

Contrast	AUC (mean ± std)	AUC gap (mean)	%sig	med n
structureType				
trivial / or_only	0.520 ± 0.048	0.043	0%	130
trivial / and_only	0.502 ± 0.040	0.033	3%	127
trivial / and_or_mixed	0.502 ± 0.042	0.034	0%	144
or_only / and_only	0.506 ± 0.034	0.027	0%	179
or_only / and_or_mixed	0.518 ± 0.035	0.032	10%	196
and_only / and_or_mixed	0.503 ± 0.040	0.032	7%	193
predicateCombo				
single / few	0.523 ± 0.035	0.033	13%	271
single / many	0.519 ± 0.048	0.041	7%	130
few / many	0.527 ± 0.048	0.045	17%	246
paramType				
none / small	0.571 ± 0.046	0.071	40%	114
none / medium	0.525 ± 0.035	0.035	3%	150
none / large	0.568 ± 0.038	0.069	33%	148
small / medium	0.530 ± 0.037	0.037	20%	175
small / large	0.504 ± 0.036	0.028	3%	173
medium / large	0.493 ± 0.027	0.022	0%	209
nestingDepth				
flat / shallow	0.522 ± 0.038	0.036	3%	102
flat / medium	0.555 ± 0.032	0.055	27%	140
flat / deep	0.468 ± 0.046	0.047	3%	160
shallow / medium	0.521 ± 0.041	0.039	7%	164
shallow / deep	0.533 ± 0.036	0.039	17%	184
medium / deep	0.563 ± 0.028	0.063	53%	221

med n: median test-set size over seeds.

6.7. Unsupervised Clustering: A Label-Free Cross-Check

The clustering attacker stated in Section 6.2 tests whether the on-chain observables group by a hidden label without access to that label. Table 12 and Figure 2 summarize the clustering results across the three privacy objectives.

ARI stays within ±0.003 for every PCH, PI, and SPP target except the SPP `nestingDepth`, indicating little agreement between the clusters and the hidden labels. The PCH and SPP NMI values are below 0.03. The highest, 0.0281 for the ICD-10 chapters, lies within the range expected without signal for ten classes of these sizes (Appendix A).

For `nestingDepth`, the ARI is small but statistically significant under the label-permutation test reported in Table 12. To investigate this association, we repeat clustering on the same executions, retaining commit gas and replacing access gas with the residual defined in Equation (19). As reported in the table note, the nesting-depth ARI then falls to near zero and is no longer statistically significant. Together with the diagnostic for the `medium/deep` contrast in Section 6.6, this result is consistent with a chance association between the randomly drawn padding targets and nesting depth in the evaluated dataset. However, the residual analysis yields a small but nominally significant association for `paramType`. These findings provide complementary evidence to the supervised results on the same dataset.

Table 12. Unsupervised clustering attack (K-Means, K = number of classes, $K = 2$ for pairs) on the on-chain gas features.

Objective	Target	n	#cls.	ARI	p_{ARI}	NMI
PCH	DUO terms	1773	10	0.0000	0.48	0.0109
PCH	ICD-10 chapters	770	10	0.0021	0.06	0.0281
PI	policy pairs (30v30)	127	2	0.0020	—	0.0139
SPP	`structureType`	1125	4	0.0004	0.33	0.0034
SPP	`predicateCombo`	1125	3	−0.0005	0.58	0.0015
SPP	`paramType`	1125	4	−0.0000	0.49	0.0030
SPP	`nestingDepth`	1125	4	0.0071	< 0.01	0.0088

PCH clusters all executions globally; SPP uses its 1,125-execution view (Section 6.3). For PI, n counts balanced pairs (Section 6.2), and the NMI shown is not debiased; after permutation debiasing, the mean NMI is 0.001, and 6 of the 127 pairs (5%) are flagged, the rate expected under the null at $\alpha = 0.05$. The p_{ARI} column reports label-permutation tests for PCH and SPP; for PI, ARI and NMI are means across the balanced policy pairs, and significance is assessed separately for each pair using the permutation-debiased NMI test. The dash (—) indicates that no p-value was computed for the mean PI ARI. Replacing access gas with the residual while retaining commit gas gives ARI −0.0004 ($p_{\text{ARI}} = 0.60$) for `nestingDepth` and ARI 0.0026 ($p_{\text{ARI}} = 0.04$) for `paramType`.

6.8. Gas-Padding Ablation

Because gas usage is the only observable that varies across the evaluated policies (Section 6.1), the ablation measures how much of the policy-dependent gas signal randomized gas padding removes.

We compare two protection configurations on the same selected policy set, differing only in whether randomized gas padding is enabled. **B1** uses the fixed circuit, commitment-only interface, and TPS execution profile, with gas padding disabled for both policy commitment and access transactions. **B2** enables randomized gas padding for both transaction types, yielding the full defense configuration used in the evaluation. The policy subset is selected deterministically from the evaluation corpus. Each content label and structural bucket is represented by at least 70 executions from at least 8 distinct policies. All policies in the depth layer (Section 6.3) are retained.

The paired leakage metric is

$$\Delta_{\text{leak},s} = |\text{AUC}_{\text{B1},s} - 0.5| - |\text{AUC}_{\text{B2},s} - 0.5|, \tag{20}$$

where positive values mean padding moved the attacker closer to chance. For each task we compute $\Delta_{\text{leak},s}$ on each of 30 paired train/test splits s, in which B1 and B2 share the same test policies, and report its mean Δ_{leak} with a 95% percentile-bootstrap confidence interval (CI) over the splits (Table 13 and Figure 3).

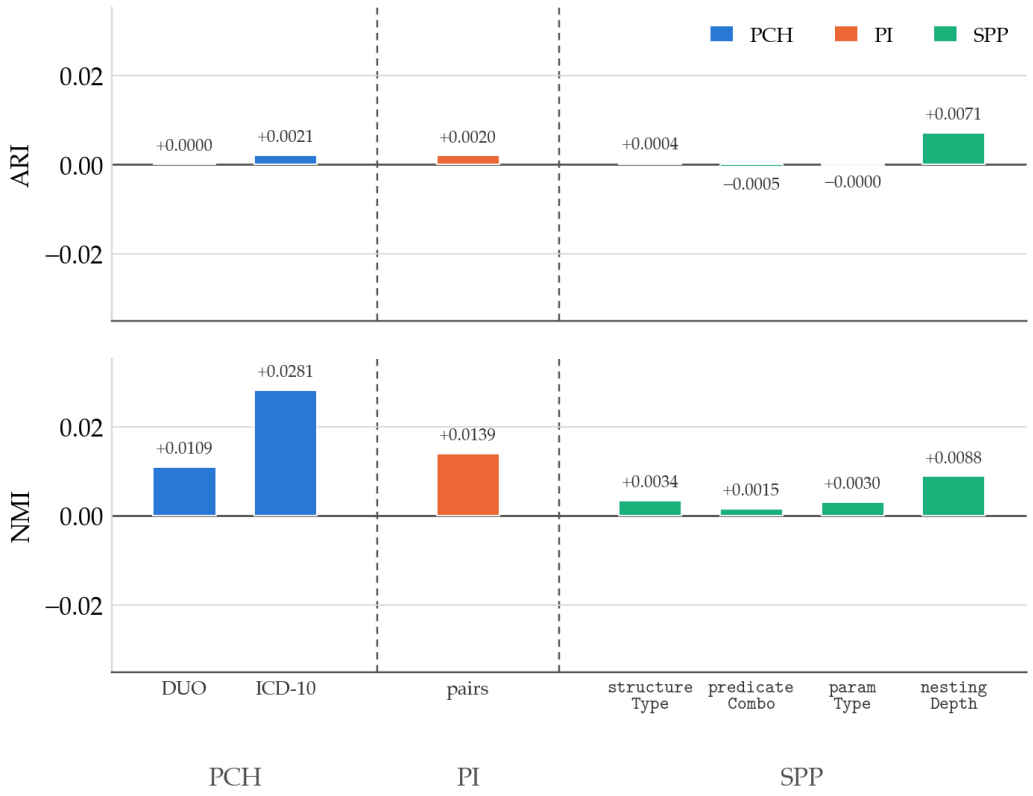


Figure 2. Unsupervised K-Means clustering across three objectives: ARI (top) and NMI (bottom) per target.

Table 13. Gas-padding ablation on Polygon Amoy: B1 (no padding) versus B2 (deployed randomized gas padding) on the same execution subset; supervised rows are the largest-signal contrasts per experiment (per axis for SPP).

Attack	Target	B1 (unpad.)	B2 (pad.)
Supervised (RF AUC; Δ_{leak} with 95% interval)			
SPP	`structureType trivial/and_or_mixed` $\Delta_{\text{leak}} = 0.347$ CI$[0.334, 0.360]$	0.902	0.490
SPP	`predicateCombo single/few` $\Delta_{\text{leak}} = 0.301$ CI$[0.287, 0.314]$	0.844	0.494
PCH	DUO HMB $\Delta_{\text{leak}} = 0.337$ CI$[0.319, 0.352]$	0.876	0.495
PI	`controlled_difference` † $\Delta_{\text{leak}} = 0.088$ CI$[0.062, 0.111]$	0.676	0.446
Unsupervised (NMI, B1−B2 with bootstrap 95% interval)			
PCH	DUO terms	0.349 [0.316, 0.407]	
SPP	logic structure	0.102 [0.018, 0.138]	
PI	balanced pairs (debiased NMI)	0.670 [0.591, 0.743]	
Gas cost (mean over the subset, B1 → B2)			
Commit gas		219,737	304,098 (+38.4%)
Access gas		723,462	1,056,674 (+46.1%)

† Positive control: the least similar pairs by construction, excluded from the PI PASS/FAIL decision.

Figure 3 plots supervised tasks by their deviation from chance, $|\mathrm{AUC} - 0.5|$, without padding (B1, horizontal axis) and with padding (B2, vertical axis); a point below the diagonal is a task whose attacker moved closer to chance once padding was applied. Without padding, 21 tasks deviate from chance by more than 0.15; with padding, all 21 fall below this level, and across all tasks the median deviation drops from 0.137 to 0.038. By family, the median Δ_{leak} is $+0.173$ for SPP and $+0.058$ for PCH. For PI it is -0.011, because three of its four tasks already deviate from chance by at most 0.08 without padding; the fourth, the positive control, drops from AUC 0.68 to 0.45.

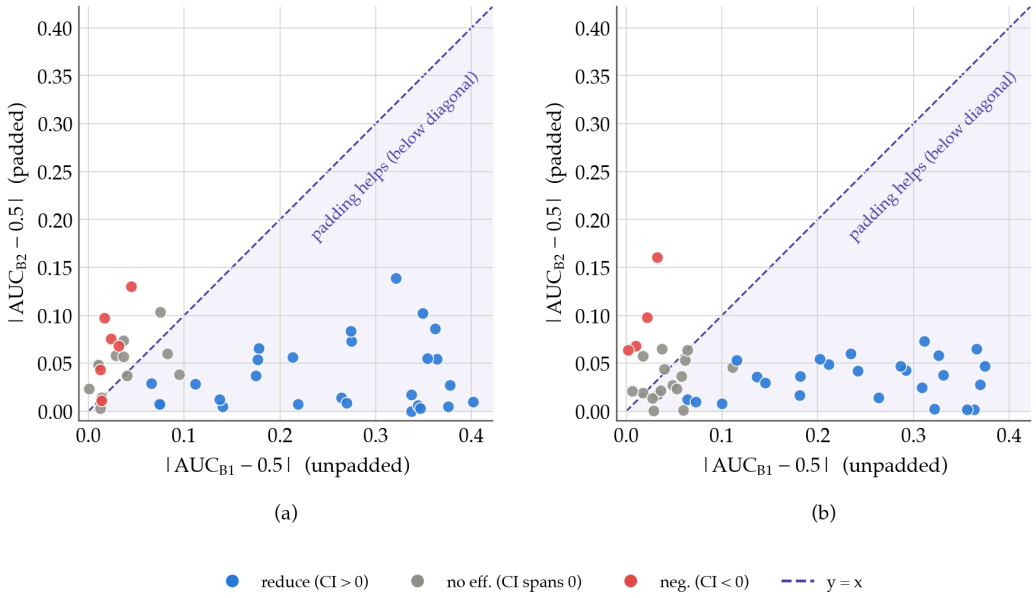


Figure 3. Gas-padding ablation over supervised tasks: (**a**) RF; (**b**) LR.

With a fixed circuit, the Groth16 verifier performs the same operations for every proof. The access-control path is also the same across the evaluated policies. The unpadded access gas of the 1,119 executions spans only 723,288–723,564 gas (one cold-start first call at 741,276 aside), in steps of 12 gas, and nine of the ten ICD-10 chapter tasks stay near chance on it (RF AUC 0.455–0.536); the exception, ch6 (RF AUC 0.713), has only 9 policies in the subset.

Decoding the calldata of the transactions traces this variation to the requester pseudonym id_{req}; the 12-gas step is the price difference between a zero and a non-zero calldata byte set by Ethereum Improvement Proposal 2028 (EIP-2028) [45]. In the evaluation, a requester who presents a profile credential is identified by a Poseidon-derived pseudonym that fills the 256-bit word, whereas a requester of an ontology-only policy carries a 128-bit random pseudonym whose upper 16 bytes are zero; the 192-gas difference between the two encodings is exactly the gap between the two clusters of B1 gas values, and whether a policy activates the profile predicate is correlated with its DUO terms, its structure type, and its parameter bucket. The remaining spread of 84 gas within each cluster comes from the zero-byte counts of the proof elements, the policy and request identifiers, and the epoch, none of which depends on the policy.

The unpadded leak is therefore a deterministic 192-gas calldata-pricing difference in how one public transaction argument is encoded, the class of channel that padding is designed to absorb. The padding routine counts the zero and non-zero calldata bytes of the transaction and tops its gas up to the drawn target, so this 192-gas difference is compensated exactly; the overhead that remains after padding varies by less than 100 gas (Section 6.4), and randomizing the target by $\pm 10\%$ ($\pm 105{,}000$ gas) masks it. Encoding every

pseudonym at full width would close this particular channel without padding; randomized gas padding remains the more general defense because it also absorbs deterministic gas differences caused by other public encodings or execution effects.

As summarized in Table 13, the clustering attacker shows the same pattern on a second, label-free attack family: the debiased NMI of the balanced PI pairs is 0.67 higher under B1 than under B2, and the DUO-term and logic-structure clusterings lose 0.35 and 0.10 of NMI, each with a policy-group bootstrap interval excluding zero.

The 12 tasks with no detectable effect sit in the lower-left of the figure, where B1 is already near chance, and the 6 tasks on which padding *raises* the deviation (negative Δ_{leak}, shown in red) are cases where B1 is at chance and the padded transcript's remaining deviation is the per-policy padding-target draw examined in Sections 6.4 and 6.6.

The TPS samples the gas-padding target uniformly from 945,000–1,155,000 gas. The lower bound exceeds the unpadded access cost measured in B1 on Amoy (approximately 723,000 gas). On average over the subset, padding adds roughly 330,000 gas per access transaction (+46%) and 85,000 gas per policy commitment (+38%; Table 13). The mean access overhead is nearly independent of policy complexity, because unpadded access costs differ by less than 300 gas across policies and all policies use the same padding band.

Table 14 gives the single-point cost characterization of the evaluated prototype configuration at $L = 64$; its prover timings were measured on the two-core capture host, and its transactions use the context-bound five-signal statement. Within these costs, binding the program length (Section 5.3) adds 88 constraints and no public input, and binding the authorization context adds 605 constraints and one public input. At Amoy's gas schedule, the context binding adds 64,022 gas per access and recording the bound context on-chain adds 68,946 gas, together 18% of the mean unpadded access cost; because the unpadded cost stays below the padding band, the binding does not change the padded cost.

Table 15 reports two offline benchmarks of circuit capacity, each with 30 verified proofs per configuration on an Apple M1 using the rapidsnark Groth16 prover. With $N = 13$, $M = 11$, and eight Merkle levels fixed, increasing L from 32 to 128 raises the circuit from 64,945 to 134,009 constraints and the median proving time from 0.411 to 0.734 s. With $L = 64$ fixed, increasing the ontology-tree depth from 4 to 16 raises the circuit from 73,876 to 97,468 constraints and the median proving time from 0.512 to 0.571 s for a five-slot ontology-heavy policy. Every variant retains five public signals, so enlarging either capacity changes the prover artifacts but not the verifier interface.

Table 16 evaluates policy content within the evaluated circuit ($L = 64$, eight Merkle levels). Across simple, ontology-heavy, profile-heavy, and max-mixed policies, median witness time is 0.335–0.354 s and median proving time is 0.484–0.545 s, despite the encoded programs ranging from 3 to 61 tokens and activating from one to all 11 production predicate slots. The larger end-to-end time of the profile-backed policies (about 0.64 s) comes from the benchmark harness, which issues and signs a fresh profile credential for every request; in deployment the issuer signs a credential once, so this cost is not incurred per request. The fixed-circuit witness, prove, and verify work remains stable. Under load, throughput stays between 0.79 and 0.94 proofs/s as concurrency rises from one to four, while median latency grows from 1.2 to 4.1 s.

7. Discussion

Privacy findings. Under the passive public-chain observer model, the evaluated attacks generally perform near chance. Two supervised tasks are classified as FAIL under the leakage criterion, and the clustering analysis detects one small signal. The diagnostics attribute these signals to the randomly sampled gas-padding targets (Sections 6.4–6.7). Such padding-induced failures are a property of the criterion on a corpus in which

Table 14. Single-point cost profile of the evaluated prototype configuration at $L = 64$. Gas, logs, calldata, and local prover timings are summarized over the 1,773 Polygon Amoy executions.

Metric	Value / source
Circuit size	81,740 constraints; 80,884 wires; $L = 64$
Public interface	4 public inputs plus 1 output
Padded on-chain gas	Commit 304,167; access 1,057,283 mean gas
Event logs	Commit: 1 log / 180 bytes; access: 2 logs / 296 bytes
Calldata length	Commit 196 bytes; access 612 bytes (constant across policies)
Local witness time	763.6 ms mean
Local prove time	2,884.2 ms mean

Table 15. Offline benchmarks of circuit capacity (30 verified proofs per row; every variant keeps $N = 13$, $M = 11$, and five public signals).

Parameter	Capacity	Constraints	Wires	WASM (MB)	zkey (MB)	Witness	Prove	Verify
Program capacity: Merkle levels fixed at 8; mixed policy								
Program length L	32	64,945	64,441	8.6	38.7	285 (343)	411 (562)	11 (16)
Program length L	64	81,740	80,884	8.7	52.2	350 (414)	521 (761)	11 (22)
Program length L	128	134,009	132,449	10.1	87.5	535 (682)	734 (984)	11 (15)
Ontology capacity: $L = 64$; five-slot ontology-heavy policy								
Merkle levels	4	73,876	73,052	8.6	48.3	352 (415)	512 (605)	11 (17)
Merkle levels	8	81,740	80,884	8.7	52.2	347 (354)	499 (539)	11 (16)
Merkle levels	12	89,604	88,716	8.7	56.0	358 (395)	552 (632)	10 (15)
Merkle levels	16	97,468	96,548	8.7	59.9	370 (456)	571 (713)	10 (13)

Timing cells report median (p95) in ms; WASM and zkey are the sizes of the WebAssembly (WASM) witness generator and the proving key, respectively.

Table 16. Policy-content and workload scalability of the evaluated circuit ($L = 64$, eight Merkle levels) on one Apple M1 host using rapidsnark.

(a) Policy content

			Time (ms): median (p95)			
Policy	Tokens	Slots	Witness	Prove	Verify	End-to-end
Simple	3	1	354 (418)	545 (746)	12 (14)	925 (1170)
Ontology-heavy	19	5	349 (391)	506 (550)	10 (12)	880 (939)
Profile-heavy	31	1	337 (349)	492 (522)	10 (11)	1479 (1516)
Max-mixed	61	11	335 (346)	484 (537)	10 (10)	1468 (1538)

(b) Load

Concurrency c	Throughput (proofs/s)	Latency (ms): median (p95)
1	0.805	1230 (1659)
2	0.787	2492 (3805)
4	0.936	4113 (5854)

Tokens: program length ℓ (at most $L = 64$); slots: external-predicate slots the policy uses (of $M = 11$). End-to-end time also includes off-circuit evidence preparation, which for the profile-backed policies includes the harness's per-request issuance of a signed credential (about 0.64 s). Latency is each request's processing time while up to c requests run at once; queueing is excluded. Five initial requests at each concurrency level are excluded from all timings.

coverage-layer policies are executed once, so the thirty seeds resample the same random draws; we keep the criterion as stated and document the diagnostic, since raising the significant-seed or AUC threshold far enough to absorb these cases would also weaken it against genuine leaks.

The commitment-only interface and fixed circuit avoid exposing policy logic directly in the public transcript. The paired ablation provides empirical evidence that randomized gas padding reduces leakage through gas consumption. Without padding, deterministic differences in calldata encoding yield attack AUCs of up to 0.90 (Section 6.8).

The cost of padding. The on-chain cost of this protection is the gas overhead of randomized gas padding (Section 6.8). With fixed padding bands, the mean overhead

for each transaction type is approximately constant, so total on-chain costs grow linearly with transaction volume (Section 6.8). Within the evaluated circuit, policy complexity has little effect on on-chain costs. All proofs require the same verification operations and use a fixed public-input format with five signals. The costs of witness generation and proving for larger circuit capacities or different policy contents are incurred off-chain (Tables 15 and 16).

The offline load benchmark in Table 16 rotates the four policy classes through 100 verified requests at each concurrency level. The best observed batch throughput is 0.936 proofs/s at concurrency four, with median and p95 request latencies of 4.113 and 5.854 s; CPU contention makes the concurrency-two result slightly lower. At this throughput, one prover on the benchmark host handles about 80,900 requests per day, or about 53,900 if a third of its capacity is held in reserve; the estimate is conservative, because half of the rotated requests include the harness's credential issuance (Section 6.8). On-chain, each padded access costs 1,057,283 gas on average (about 0.032 POL, the native token of Polygon, at the 30 gwei tip of the evaluation, against 0.022 POL unpadded; Table 14), and each policy commit adds a one-time 304,167 gas (0.009 POL).

The padding band must be positioned above the unpadded access cost on the target network, which differs between networks because the verifier's cost does (243,000 gas on a local EVM and 372,000 on Amoy, whose BN254 precompiles are priced higher); a deployment therefore calibrates the center of the band once per network, and its width, $\pm 10\%$ here, determines how large a residual difference the random target can mask.

TPS trust and credential provenance. The TPS trust assumption is the main architectural tradeoff. Moving confidentiality-sensitive proof generation to the TPS keeps policy-bound proving material away from requesters and public smart contracts while preserving on-chain verifiability, but it shifts part of the confidentiality burden to the service layer (Section 4). It also affects liveness and accountability. The TPS can refuse to generate a proof, and the on-chain record alone may not reveal the reason for that refusal.

Attribute origin is only partially delegated to the TPS. For the profile-backed DUO terms, a compromised TPS cannot alter the signed claims or the fields bound with them (Sections 4.1 and 5.4). However, because the TPS receives the holder secret and credential witness in the current workflow, holder binding does not protect against a malicious TPS reusing otherwise valid evidence within its signed scope.

Issuer governance and deployment path. The remaining trust anchor is therefore issuer governance. The prototype binds the proof to a policy-selected issuer-key commitment, but it has no independent registry establishing that this key belongs to an accredited healthcare authority, and it does not check credential revocation.

A deployment could add issuer governance without modifying the Boolean interpreter. A governed registry would publish issuer keys together with their associated credential schemas and validity epochs. Each issuer would sign a domain-separated credential containing a holder-key commitment, claim vector, authorization scope, validity interval, and unique credential identifier. The circuit would verify the credential signature and a Merkle proof that the issuer key is active in the registry, while continuing to derive policy attributes from the signed claims. The holder would authorize each request by signing the context digest and request epoch, preventing a TPS that observes one credential presentation from generating proofs for fresh sessions; an epoch-specific sparse-Merkle or accumulator root would provide a private non-revocation witness for the credential identifier.

The registry and revocation roots can occupy fixed-format public or committed parameter slots, preserving the Boolean interpreter and the shape of the policy-independent public interface, although these checks require a revised circuit and new proving and ver-

ification keys. Existing healthcare credentials using Ed25519 or the Elliptic Curve Digital Signature Algorithm require either higher-cost in-circuit verification or an additional SNARK-friendly credential; TPS re-signing cannot establish issuer-origin assurance.

Policy and context binding. The policy commitment binds the active program length ℓ, and the circuit ties the public digest d_χ to the requester pseudonym, resource, and nonce (Section 5.3); an accepting proof therefore can neither skip part of the committed policy nor be accepted under another context. This context-binding guarantee applies within a single ACM deployment. The digest does not bind the transaction submitter, chain identifier, or contract address. Additional checks are therefore needed to prevent another party from copying and submitting the same request first, or reusing the proof on another chain or contract.

Future directions. Based on the TPS trust assumption, its effect on accountability, and the deployment scope of context binding discussed above, we identify several directions for future work. For example, adding an explicit chain/contract domain and an authenticated submission channel would extend context binding beyond a single deployment and reduce front-running risk. In addition, accountable proof obligations, such as on-chain commitments to serve requests combined with timeout-with-evidence mechanisms, would make proof refusals attributable. Finally, the reduced-trust TPS instantiations described in Section 4, based on collaborative SNARKs or TEE-backed proving, would relax the trust concentration itself, including the partial delegation of attribute origin.

8. Conclusion

This paper presented TPS-ZKPI, a Trusted Proof Service-assisted zero-knowledge approach to policy-private consent enforcement on public blockchains. In TPS-ZKPI, the data provider commits to a hidden consent policy on-chain, the TPS generates a zero-knowledge proof of each access decision under this policy, and a smart contract verifies the proof. Each policy is compiled into a fixed-length program for a reusable zero-knowledge policy interpreter, so all policies share one verification interface, and randomized gas padding masks the remaining gas differences. We evaluated TPS-ZKPI on the Polygon Amoy testnet with supervised and unsupervised attacks in which a passive public-chain observer tried to infer hidden policy properties from public-chain metadata. The results showed no leakage under our criterion in 43 of the 45 supervised tasks; the other two traced to chance correlations with the padding, and unsupervised clustering found no policy structure beyond the same padding effect. Furthermore, the gas-padding ablation showed that, without padding, the same attacks often recovered policy content and structure, which confirmed that padding protects the gas channel at a fixed gas cost per transaction. Overall, this paper showed empirically that TPS-ZKPI resists the modeled metadata attacks and provides auditable consent enforcement on a public blockchain.

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Abbreviations

The following abbreviations are used in this manuscript:

ABAC	Attribute-Based Access Control
ABE	Attribute-Based Encryption
ACM	Access-Control Manager
ARI	Adjusted Rand Index
AUC	Area Under the Receiver Operating Characteristic Curve
CI	Confidence Interval
CP-ABE	Ciphertext-Policy Attribute-Based Encryption
DUO	Data Use Ontology
ECDSA	Elliptic Curve Digital Signature Algorithm
EdDSA	Edwards-curve Digital Signature Algorithm
EIP	Ethereum Improvement Proposal
EVM	Ethereum Virtual Machine
GA4GH	Global Alliance for Genomics and Health
HANCESTRO	Human Ancestry Ontology
HE	Homomorphic Encryption
ICD-10	International Classification of Diseases, 10th Revision
IoMT	Internet of Medical Things
LR	Logistic Regression
MONDO	Mondo Disease Ontology
MPC	Multi-Party Computation
NMI	Normalized Mutual Information
OBI	Ontology for Biomedical Investigations
PCH	Policy Content Hiding
PI	Policy Indistinguishability
PRE	Proxy Re-Encryption
RF	Random Forest
RPN	Reverse Polish Notation
SAML	Security Assertion Markup Language
SNARK	Succinct Non-interactive Argument of Knowledge
SNOMED CT	SNOMED Clinical Terms
SPP	Structure and Parameter Privacy
SSI	Self-Sovereign Identity
TEE	Trusted Execution Environment
TPS	Trusted Proof Service
TPS-ZKPI	Trusted Proof Service-assisted Zero-Knowledge Policy Interpreter
WASM	WebAssembly
XACML	eXtensible Access Control Markup Language
zkVM	Zero-Knowledge Virtual Machine

Appendix A. Sample-Size Sensitivity of NMI

The clustering cross-check reads ARI directly but debiases the NMI of the balanced PI pairs (Section 6.2). To examine that choice, we quantify how *finite-sample effects* alone can inflate NMI even when there is *no* discriminative signal, using an idealized null in which all classes are statistically indistinguishable.

We sample every class from the *same* distribution, $z_i \sim \mathcal{N}(0, I_d)$ for all classes, where I_d is the d-dimensional identity covariance and $d = 2$ matches the two gas features the clustering uses. Since no class carries class-specific information, a clustering algorithm should not recover class membership, and both NMI and ARI should concentrate around 0 as the sample size grows.

For each configuration we run $T = 1000$ independent trials. In each trial we (i) generate G classes with n i.i.d. samples per class from $\mathcal{N}(0, I_d)$, for $n_{\text{tot}} = Gn$ samples in total, assigning each sample a class label by construction order (these labels are independent of the sample values, since all classes share one distribution), (ii) standardize the pooled data and apply K-Means with $K = G$ and ten initializations, as in the evaluation, and (iii) score the predicted assignments against these assigned class labels with NMI and ARI. We vary n while holding G fixed, and report the trial means $\overline{\text{NMI}}$ and $\overline{\text{ARI}}$. We consider a **PI-like** binary structure ($G = 2$, $n \in \{3, 10, 50, 100\}$), a five-class structure ($G = 5$, $n \in \{3, 10, 100\}$), and a ten-class structure ($G = 10$, $n \in \{3, 10, 77\}$) together with the observed ICD-10 chapter sizes (49–135 executions per chapter, $n_{\text{tot}} = 770$).

Table A1 shows that, although all classes are indistinguishable by construction, **NMI is noticeably positive at very small n but collapses toward zero as n grows**, whereas **ARI stays near zero throughout**. For the binary structure, $\overline{\text{NMI}}$ falls from 0.2074 at $n = 3$ ($n_{\text{tot}} = 6$) to 0.0420 at $n = 10$, and to 0.0072/0.0034 at $n = 50/100$. The same trend holds with more classes, at a higher level for every n: with five classes $\overline{\text{NMI}}$ goes from 0.4152 ($n = 3$) to 0.0102 ($n = 100$), and with ten classes from 0.5346 ($n = 3$) to 0.0239 ($n = 77$). For the observed ICD-10 chapter sizes, a separate null run that averages each trial over 20 K-Means seeds, as the evaluation does, gives a null NMI with mean 0.0246 and 95th percentile 0.0302; the observed ICD-10 NMI of 0.0281 is reached in 145 of 1000 null trials ($p = 0.15$).

Table A1. Sample-size sensitivity of NMI and ARI under the null hypothesis where all classes are i.i.d. sampled from $\mathcal{N}(0, I_d)$.

n (per class)	n_{tot}	$\overline{\text{NMI}}$	$\overline{\text{ARI}}$
Two classes (as in a PI pair)			
3	6	0.2074	0.0127
10	20	0.0420	0.0008
50	100	0.0072	−0.0003
100	200	0.0034	−0.0004
Five classes			
3	15	0.4152	−0.0034
10	50	0.1226	0.0003
100	500	0.0102	−0.0000
Ten classes (as the DUO and ICD-10 targets)			
3	30	0.5346	−0.0009
10	100	0.2087	0.0003
77	770	0.0239	−0.0001
Ten classes, observed ICD-10 chapter sizes			
49–135	770	0.0244	−0.0001

$T = 1000$ trials per row, one K-Means fit with ten initializations per trial; n_{tot} is the total sample size. For the observed ICD-10 chapter sizes, the text also reports a separate run that averages each trial over 20 K-Means seeds, as the evaluation does.

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